

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Second Semester MCA (2 Year) Degree Regular and Supplementary Examination June 2023

Course Code: 20MCA188**Course Name: ARTIFICIAL INTELLIGENCE**

Max. Marks: 60

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

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| 1 | What is Knowledge? How is it classified? | (3) |
| 2 | What is meant by state space representation of a problem? Explain. | (3) |
| 3 | What are admissible Heuristics? Give examples. | (3) |
| 4 | What is Steepest ascend hill climbing? How does it differ from Hill climbing?
Explain with appropriate examples. | (3) |
| 5 | What is the drawback of Minimax algorithm? Discuss. | (3) |
| 6 | Consider the following facts and represent in FOPL:
a) Bill has at least one sister.
b) All purple mushrooms are poisonous.
c) Every gardener likes the Sun. | (3) |
| 7 | What is the advantage of Frame based knowledge representation scheme over
FOPL? Explain with an example. | (3) |
| 8 | Compare Supervised, Unsupervised and Reinforcement learning techniques. | (3) |
| 9 | What is MYCIN? Explain. | (3) |
| 10 | What is the difference between Fuzzy Logic and Binary logic? Give examples. | (3) |

PART B*Answer any one question from each module. Each question carries 6 marks.***Module I**

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| 11 | What is a Production System in AI? Explain various types of Production
Systems with appropriate examples. | (6) |
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OR

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| 12 | Explain various task domains of AI. | (6) |
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Module II

- 13 Compare Best first search and Breath first Search algorithms. (6)

OR

- 14 Illustrate Iterative Deepening search with an example. (6)

Module III

- 15 Discuss Alpha-Beta cut-off algorithm with the help of an example. (6)

OR

- 16 Express the following concepts as a semantic network structure with interconnected nodes and labeled arcs: (6)

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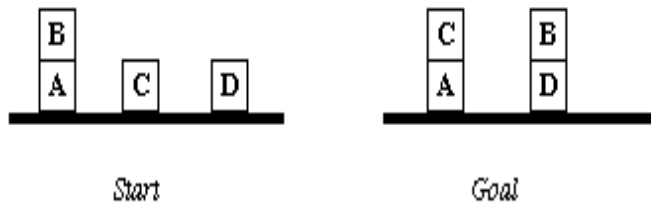
Module IV

- 17 Consider the following facts and represent it in FOPL: (6)

Ram is a good student. All good students have high grades. All good students with high grades are bright. Show that Ram is bright using Resolution.

OR

- 18 What is Goal Stack planning? Solve the following Blocks World problem using goal stack planning algorithm. (6)

**Module V**

- 19 What is the importance of an Expert System? Mention its characteristic features. (6)

Explain Architecture of an expert system.

OR

- 20 With appropriate examples, perform any three operations on Fuzzy sets? (6)
